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Prof. Jose Manuel Amigo
Editor, Chemometrics and Intelligent Laboratory Systems

Revised manuscript CHEMOLAB-D-26-01046

Dear Prof. Amigo,

Please find enclosed the revised version of our manuscript, together with a point-by-point response to Reviewer 2 and a version with all changes marked.

All four comments of the reviewer have been addressed. In addition, and on our own initiative, we corrected the predictive comparison with partial least squares: the number of PLS latent variables is now selected by cross-validation, and train/test splits keep replicate spectra of the same sample together. This changes the conclusion of hypothesis H1 in the near-infrared branch (no advantage over calibrated PLS), while confirming it in the Raman branch; the gate-topology and prior-stability results are unaffected. Because of this change we propose a new title:

Gate Topology and Reproducibility of Prior-Guided Spectral Gating Across Near-Infrared and Raman Spectroscopy

The response document explains these changes in detail. We recognise that they go beyond a minor revision and would welcome further review if you consider it appropriate. Code, data identifiers, and all revision results are available at <https://github.com/clarimar/pgsg2-nir-raman-gating>.

Both authors have approved the revised manuscript. The authors declare no conflicts of interest.

Sincerely,
Clarimar José Coelho, on behalf of both authors